COMPARATIVE ANALYSIS

FraudDNA: An Adaptive History-Aware Framework for Financial Transaction Fraud Detection

---

Course: Applied Machine Learning

Primary source: ALT_Paper.pdf

Format reference: Team10-AI_Code_Review_Complete_Comparative_Analysis.pdf

Scope and evaluation note
This report uses the experimental results explicitly reported in the supplied FraudDNA paper. It separates measured comparisons within FraudDNA from qualitative comparisons with research families discussed in the paper. No external paper metrics have been invented or treated as directly comparable.

Executive Summary

FraudDNA addresses a practical limitation in financial fraud detection: usable behavioral history is not available for every transaction. Rather than forcing one history-enriched model to score all transactions, the framework routes each transaction to a transaction-only model when destination history is absent and to a history-enriched model when prior destination transactions are available. A deterministic WHY-NOW layer explains the route and its decision context.

On the paper's chronological PaySim test set, Adaptive FraudDNA achieved the highest reported PR-AUC (0.7549), compared with 0.7252 for Model A and 0.7160 for Global Model B. It also recorded the highest reported ROC-AUC (0.9716), precision (0.0924), and F1-score (0.1671) among the three configurations. Model A had the highest recall (0.8812). These results support the adaptive-routing hypothesis for this experimental setup, not universal superiority across datasets.

1. Project Overview and Experimental Design

Dataset and split

The study uses PaySim, a synthetic mobile-money transaction dataset with approximately 6.36 million transactions. The paper reports a chronological split rather than random shuffling, to better reflect future transactions being scored using only information available before the decision.

Partition	Transactions	Fraud cases
Training	6,064,036	5,621
Testing	298,584	2,592

Model configurations

Configuration	Inputs / routing	Purpose in comparison
Model A	Nine transaction-level features; no destination-history features.	Transaction-only baseline and fallback for cold-start cases.
Global Model B	Transaction features plus destination-history features, applied globally.	Tests whether adding history for every transaction helps.
Adaptive FraudDNA	Uses Model A if valid destination history is absent; Model B when history exists.	Tests whether history should be used conditionally.

> Leakage control: the paper states that historical features use transactions preceding the current transaction, and the state update occurs after prediction. This ordering is important for avoiding temporal leakage.

2. Qualitative Comparative Analysis

The following table summarizes the approach families discussed in the source paper. It is a qualitative synthesis of the paper's literature discussion, not a claim that every cited study has the same implementation or limitations.

Approach family	Typical strength described in the paper	Key limitation for the problem	How FraudDNA differs
Transaction-level supervised ML	Uses transaction attributes and engineered features; provides a practical baseline.	Cannot directly use entity behavior that is absent from the current transaction record.	Retains a transaction-only pathway for cases without usable history.
Behavioral / aggregation models	Summarize patterns across previous transactions and can improve fraud discrimination.	Assume useful history is available; historical features must be constructed without future information.	Checks history availability at inference and uses only prior transactions.
Sequential / temporal models	Represent dependencies across transaction sequences.	Require careful temporal ordering; future information can leak into features if constructed incorrectly.	Uses a chronological holdout and post-prediction state update.
Graph neural network methods	Model relations between entities and can capture patterns not explicit in individual records.	Graph construction, dynamic updates, and temporal leakage require care; graph modeling is not the central method evaluated in FraudDNA.	Uses destination-level history routing rather than claiming a GNN architecture.
Explainable-AI methods	Can help reviewers understand model outputs through attribution or local explanations.	Some explanation methods may not directly explain why a particular inference pathway was selected.	WHY-NOW gives deterministic decision-context explanations; it does not attribute internal neural-network reasoning.
FraudDNA	Combines transaction-only and history-enriched neural pathways with availability-based routing.	Evaluated primarily on synthetic PaySim and specific model/feature configurations.	Its contribution is explicit history-availability routing and a deterministic explanation of that routing.

Interpretation

FraudDNA's main distinction is not simply adding historical features. Its central design choice is to treat history as conditional context. The global history-aware baseline underperformed the transaction-only baseline on PR-AUC, whereas adaptive routing achieved the best PR-AUC among the three tested configurations.

3. Numerical Comparative Analysis

All values in the table below are transcribed from the FraudDNA paper's reported overall chronological test results. Metrics are shown on their reported 0–1 scale.

Model	PR-AUC	ROC-AUC	Precision	Recall	F1-score
Model A	0.7252	0.9644	0.0617	0.8812	0.1153
Global Model B	0.7160	0.9701	0.0907	0.8603	0.1641
Adaptive FraudDNA	0.7549	0.9716	0.0924	0.8669	0.1671

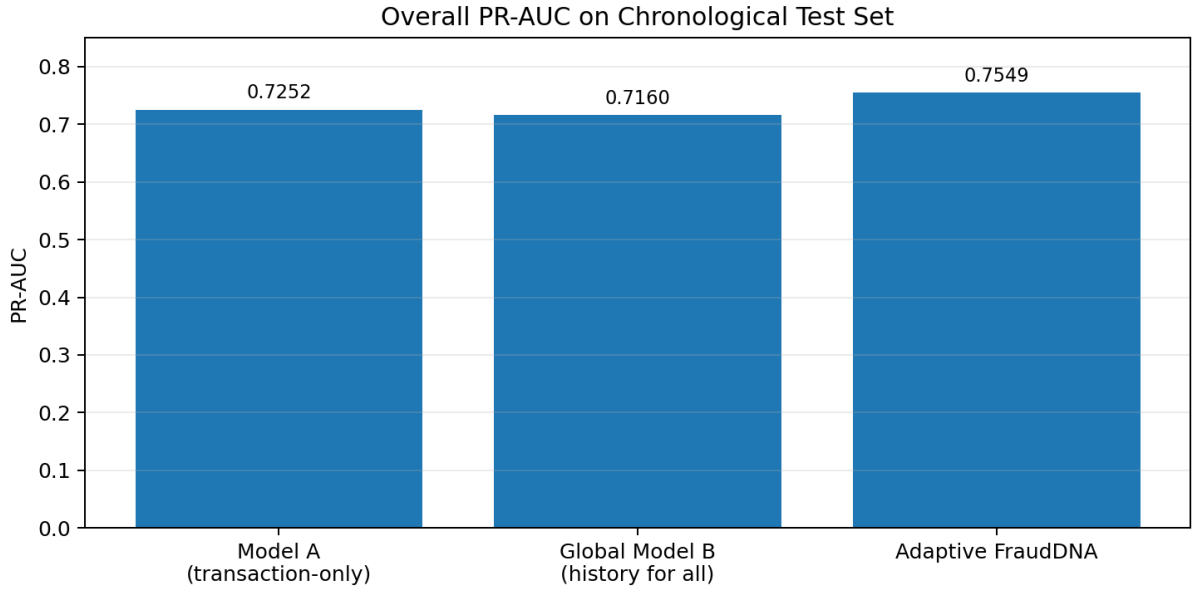


Figure 1. PR-AUC comparison across the three FraudDNA configurations. Adaptive FraudDNA has the highest reported PR-AUC.

Key findings

PR-AUC: Adaptive FraudDNA improves on Model A by 0.0297 and on Global Model B by 0.0389. The global history-aware model is 0.0092 below Model A, showing that applying history features universally did not help this metric.

ROC-AUC: Adaptive FraudDNA records 0.9716, narrowly above Global Model B (0.9701) and Model A (0.9644). ROC-AUC is high for all three configurations; PR-AUC is especially relevant here because fraud is a rare class.

Precision and F1: Adaptive FraudDNA records the highest precision (0.0924) and F1-score (0.1671) in the reported table. Absolute precision remains low, so the system would still produce many false alerts at the evaluated operating point.

Recall: Model A has the highest recall (0.8812), compared with 0.8669 for Adaptive FraudDNA. Adaptive routing therefore does not maximize every metric; its reported advantage is strongest in PR-AUC and the balance reflected by F1.

4. Multi-Metric and Conditional Performance

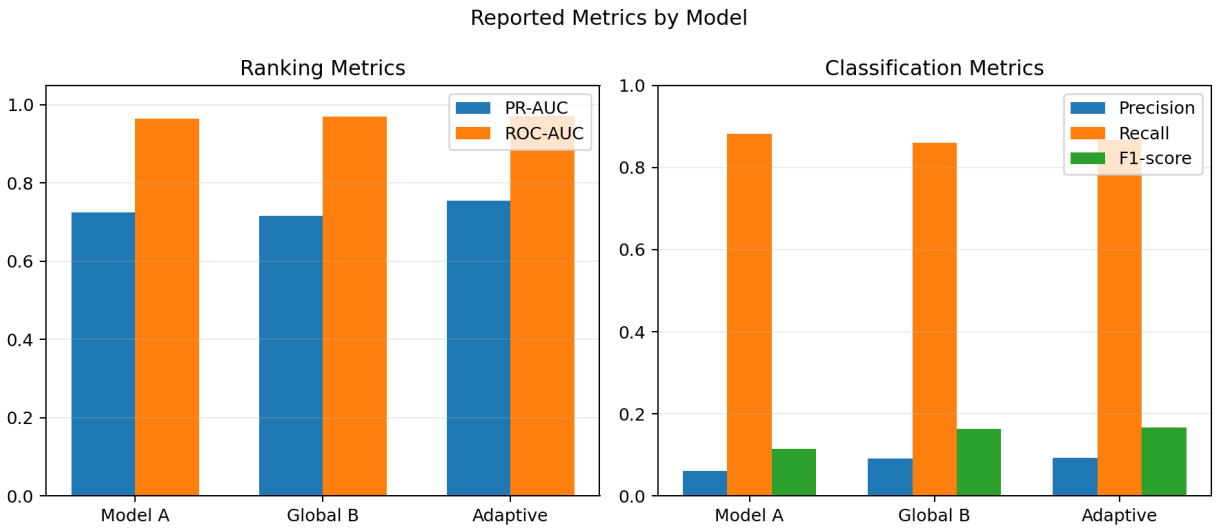


Figure 2. Reported ranking and classification metrics. The two panels use separate metric groupings for readability.

Conditional PR-AUC by history availability

The paper reports separate results for transactions whose destination history is available and cold-start transactions. These conditional values help explain why routing can be preferable to applying the history-enriched model universally.

Test group	Model A PR-AUC	Contextual Model B PR-AUC	Difference (B – A)
History available	0.3975	0.5316	+0.1341
Cold start	0.9068	0.8847	–0.0221

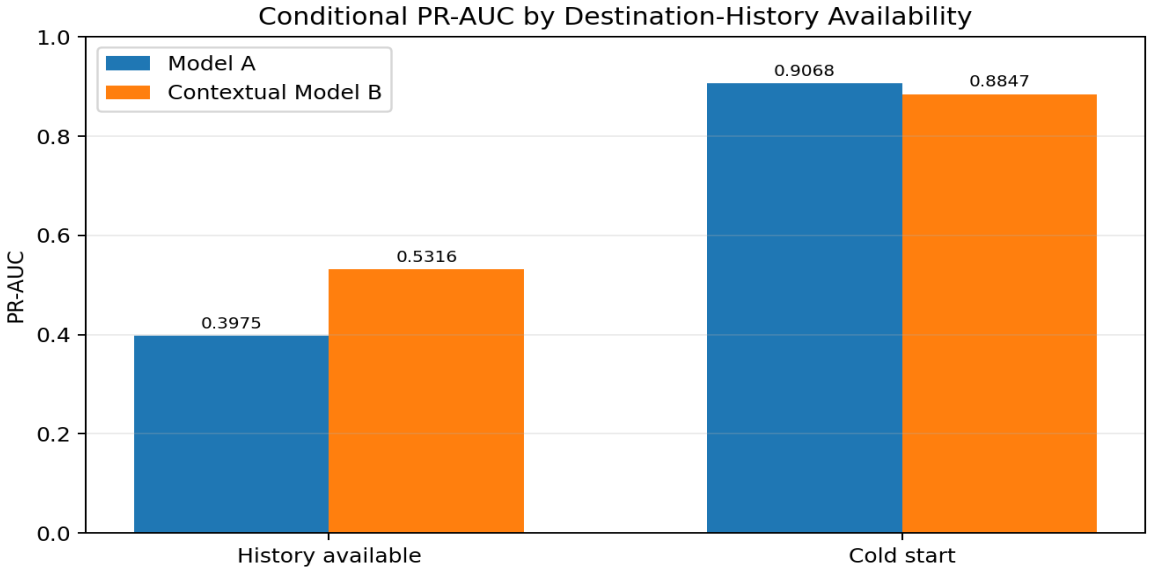


Figure 3. Conditional PR-AUC by destination-history availability. Model B performs better in the history-available group; Model A performs better in the cold-start group.

This is the central empirical justification for adaptive routing: history-enriched features help when history exists, but can be less suitable when it does not. Adaptive FraudDNA routes to the model suited to the observed history condition.

5. Stability and Explainability Analysis

Multi-seed stability

The paper reports Adaptive FraudDNA PR-AUC across three seeds. The reported mean is 0.7542 and standard deviation is 0.0029, indicating limited variation across these three runs. This is a small stability check, not a substitute for repeated experiments across datasets or time periods.

Seed	Adaptive FraudDNA PR-AUC
42	0.7572
123	0.7550
2026	0.7503
Mean	0.7542
Standard deviation	0.0029

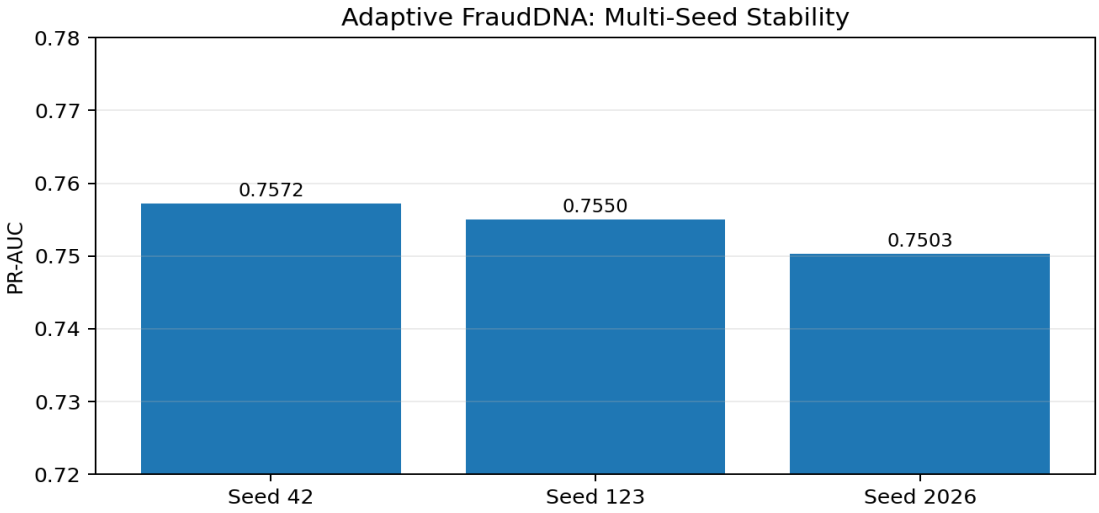


Figure 4. Adaptive FraudDNA PR-AUC across the three reported seeds.

WHY-NOW explanation layer

The paper describes deterministic explanations that report the inference route, whether destination history was available, and relevant context such as historical transaction count and average amount. For cold-start transactions, the explanation indicates that Model A was selected because destination history was unavailable. For history-available transactions, it explains why the contextual pathway was used.

Important boundary: WHY-NOW explains decision context and route selection. The paper explicitly does not claim that it explains internal neural-network feature attribution or the model's internal reasoning.

6. Research Gaps and Project Contribution

Observed issue in the problem setting	FraudDNA's response	Evidence / boundary
Usable entity history may be absent for new or rarely observed entities.	Routes to a transaction-only model when destination history is unavailable.	Explicit rule: Model A if history count is zero; Model B if history exists.
History features may not help every transaction.	Uses the history-enriched pathway conditionally rather than globally.	Global Model B PR-AUC is 0.7160 vs Model A 0.7252; adaptive PR-AUC is 0.7549.
Temporal leakage can inflate results if future transactions enter historical features.	Builds history from prior transactions and updates state after prediction.	Chronological train/test split; no future transactions used for current history.
Model routing can be opaque to an operator.	Provides deterministic WHY-NOW decision-context output.	Explains the route and available context, not internal feature attribution.
Fraud behavior and data characteristics vary across settings.	Provides an architecture that can be tested in additional contexts.	Current evaluation uses synthetic PaySim; broader validation remains future work.

7. Limitations and Fair Interpretation

The source paper identifies several limitations that should remain visible in the comparative analysis:

• **Dataset realism:** PaySim is synthetic mobile-money data and may not represent real bank or card-payment environments.
• **History choice:** Destination history was more reusable in this dataset; this should not be generalized to every transaction domain.
• **Explanation scope:** WHY-NOW is deterministic decision-context explanation, not neural feature attribution.
• **Model scope:** The reported results apply to the specific model configurations and feature sets tested.
• **Literature comparison:** The literature review does not prove that no comparable architecture exists elsewhere.

Fair-comparison note

The numerical comparison in this report is a direct comparison among the three FraudDNA configurations evaluated on the same chronological test setup. The cited literature approaches are compared qualitatively because the supplied paper does not provide a common benchmark table with aligned datasets, splits, labels, and metric definitions for those studies. Their published results should not be ranked against FraudDNA's PaySim scores without a matched experiment.

8. Comparative Discussion

The results support the idea that historical context should be treated as a resource that is useful when available, rather than an assumption that applies to every transaction. Global Model B adds history features to all transactions but records a lower PR-AUC than Model A. Adaptive FraudDNA achieves the best overall PR-AUC by selecting the contextual pathway only when destination history exists, while retaining Model A for cold-start transactions.

The improvement is not uniform across all metrics: Model A has higher recall, while Adaptive FraudDNA has higher PR-AUC, ROC-AUC, precision, and F1-score. This distinction matters for applied fraud detection, where missing a fraud case and generating false alerts have different operational costs. The paper's results support further evaluation of adaptive routing, but do not establish universal superiority over graph neural networks or other published fraud detectors.

9. Conclusion and Future Work

FraudDNA presents an adaptive history-aware framework with two neural prediction pathways, an inference-time router based on destination-history availability, leakage-aware temporal feature construction, and deterministic WHY-NOW explanations. On the reported chronological PaySim test set, Adaptive FraudDNA achieved PR-AUC 0.7549 and ROC-AUC 0.9716, compared with PR-AUC 0.7252 for Model A and 0.7160 for Global Model B. Its three-seed mean PR-AUC was 0.7542 with standard deviation 0.0029.

Future work identified by the paper includes evaluation on additional real-world datasets, richer temporal behavioral representations, learned or uncertainty-aware routing, and more sophisticated explanation methods. For a stronger applied-machine-learning comparison, future experiments should preserve chronological evaluation and report PR-AUC, precision, recall, F1-score, computational cost, and alert-level usefulness on the same datasets and splits.

10. Reference Basis

Primary project source: *FraudDNA: An Adaptive History-Aware Framework for Financial Transaction Fraud Detection*, supplied as ALT_Paper.pdf. The numerical results and limitations in this report are drawn from that paper, particularly its experimental setup and results sections (printed pages 4–7).

Format reference: *Team10-AI_Code_Review_Complete_Comparative_Analysis.pdf*, used for the structure of qualitative comparison, numerical analysis, research-gap mapping, discussion, and conclusion.

Selected literature references as listed in the source paper include Lopez-Rojas et al. (2016), “PaySim: A financial mobile money simulator for fraud detection”; Dou et al. (2020), work on graph neural network-based fraud detection; Liu et al. (2020), “GraphConsis”; Liu et al. (2021), “Pick and Choose: A GNN-based Imbalanced Learning Approach for Fraud Detection”; and Motie and Raahemi (2024), “Financial fraud detection using graph neural networks: A systematic review.” Full bibliographic details should be taken from ALT_Paper.pdf.

Prepared from the supplied project paper and comparative-analysis reference. All numerical metrics are reported values from ALT_Paper.pdf; charts visualize those values without adding new model results.